

Overview of ELOQUENT 2024 — shared tasks for evaluating generative language model quality

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Abstract. ELOQUENT is a set of shared tasks for evaluating the quality and usefulness of generative language models. ELOQUENT aims to apply high-level quality criteria, grounded in experiences from deploying models in real-life tasks, and to formulate tests for those criteria, preferably implemented to require minimal human assessment effort and in a multilingual setting. The tasks for the first year of ELOQUENT were (1) Topical quiz, in which language models are probed for topical competence; (2) HalluciGen, in which we assessed the ability of models to generate and detect hallucinations; (3) Robustness, in which we assessed the robustness and consistency of a model output given variation in the input prompts; and (4) Voight-Kampff, run in partnership with the PAN lab, with the aim of discovering whether it is possible to automatically distinguish human-generated text from machine-generated text. This first year of experimentation has shown — as expected — that using self-assessment with models judging models is feasible, but not entirely straight-forward, and that a judicious comparison with human assessment and application context is necessary to be able to trust self-assessed quality judgments.

Keywords: Generative language models · LLM · Shared task · Self-assessed quality · Evaluation

1 Introduction

Generative language models (“LLMs”) as a foundational component in an information system are able to handle a broad variety of input data robustly and elegantly, and are able to provide appropriately creative generated output to fit a broad range of application situations and the preferences of a diverse user population. An information service with a generative language model can be built to provide a flexible low threshold conversational interface for its users: there is considerable interest to put generative language models to use in productive practical applications, across domains, sectors of society, languages, and cultural areas.

The ELOQUENT lab is intended to probe the quality of a generative language model, and to do this by addressing specifically such quality issues that are raised at the deployment time when a model is included in a system for productive downstream tasks. The lab also intends to explore the reliability of system self-assessment of model quality using other models or even the same model, and to reduce the dependence of human-assessed gold standard data sets. Using language models for assessment purposes is currently being explored for e.g. relevance judgements [10], and we expect

to see strides taken in this direction, to explore what, if any, systematic differences can be found between automatic and human quality assessments.

This first year of the ELOQUENT lab for evaluating generative language model quality, we present four experimental tasks. To test (a) *topical competence*, we have defined the *Topical Quiz* task for models to self-assess their knowledge of various topics. The focus of the task is to assess the reliability of such self-testing. To test (b) *model hallucinations*, we defined the *HalluciGen* task to explore if and how models can detect and even generate hallucinated paraphrases and translations. To test (c) *consistency of output* in face of semantically equivalent but stylistically varied input, we defined the *Robustness* task where the similarity of the output for similar prompts is assessed. To test (d) *detection capacity of machine-generated* (as opposed to human-authored) text we defined the *Voight-Kampff* task where participating models were used to generate texts which were submitted to the PAN lab for classification as human-authored vs machine-generated.

These four quality characteristics are at the forefront of current discussions of reliability and trustworthiness of generative language models and the systems built to make use of such models. The four ELOQUENT tasks are all designed to be assessable using generative models, to be straightforward and simple to execute, and to require little human assessment effort.

ELOQUENT received 55 registration sign-ups for teams to participate in various subsets of the four tasks. Of these, 8 teams submitted experiment results. This is a high attrition rate, and we will poll the participants to find out what might increase the likelihood of participants completing and submitting experiments.

2 Task 1: Topical quiz

A generative language model in practical application will in most envisioned use cases be expected to stay within given task-appropriate topical boundaries, to generate material restricted to the domain it is employed to work within, and to have competence in the terminology and conventions of that domain. Examples of relevant topical domains could be business domains, such as finance [29] or healthcare [26], or even recreational activities such as sailing or basketball, ranging to differences in how a topic is treated differently across linguistic and cultural areas or in specific demographic groups.

The topical quiz task intends to answer to the need for verifying a model’s understanding of an application domain of interest. The task is defined for a system to generate a topical quiz for some given topic; to respond to such quizzes, including the one it has generated itself; and to score responses to quizzes numerically from 1 to 10. Every participating team was given a list of topics, shared as a JSON structure, and asked to use their system or systems to generate a set of questions for each topic. The dataset includes a suggested prompt string, but participants were free to reformulate the string to fit their model or system. The generated questions were submitted in a prescribed JSON structure by the participants through a submission form. These question structures were shared back to the participants for them to use their systems to generate responses to the questions. The generated responses were then again submitted in a prescribed JSON structure by the participants using a submission form. These responses were then scored 1-10 by four systems: Reindeer-Poro, Reindeer-Mistral, GPT-SW3, and GPT-4o. An example topic with responses and scores is shown in Figure 2.

The task had 27 registered participant teams. Three teams submitted quizzes, with two teams submitting responses. The teams used Poro [16], Mistral [12] (for team "Reindeer" [20]), GPT-SW3 [9], and a modular RAG approach [11] (for team Verbanex). This rather limited set of submitted

Topic: Wine and terroir

Question (Reindeer-Poro):

What factors contribute to the unique terroir of a vineyard or region?

Response (Reindeer-Mistral): The factors that contribute to the unique terroir of a vineyard or region include the soil composition, drainage patterns, exposure to sunlight, temperature and rainfall patterns, and the presence of natural features such as mountains, rivers, and bodies of water. These factors can vary greatly from one vineyard to another, even within the same region, and can have a significant impact on the taste and character of the wine produced.

Response (Reindeer-Poro): Factors that contribute to the unique terroir of a vineyard or region include the soil type and composition, the climate, the topography, and the local microclimate. These factors can interact to influence the growth and development of the vines, as well as the flavors and characteristics of the wine.

Response (GPT-SW3): A unique terroir is the combination of natural conditions like soil composition, elevation, climate, and exposure to sunlight, as well as human factors such as farming methods, viticultural practices, and winemaking techniques that all contribute to creating a distinct quality in wine. The specific characteristics of each terroir are what make it distinctive, allowing for wines from one region to taste different than wines from another region with similar conditions.

	Reindeer-Mistral	Reindeer Poro	GPT-SW3	GPT-4o
Reindeer-Mistral	9	7	8	8
Reindeer-Poro	8	8	9	7.5
GPT-SW3	9	10	10	8

Fig. 1. A sample topic for the Topical Quiz task and a sample quiz question by Reindeer-Poro; responses by Reindeer-Poro, Reindeer-Mistral, and GPT-SW3; and scores for the responses as given by Reindeer-Poro, Reindeer-Mistral, GPT-SW3, and GPT-4o

data does not provide us with support to use the originally planned fairly elaborate scoring scheme, comparing the correlation of scoring across systems and the reliability of the self-scoring of a system to scores others do.

The quizzes generated by the systems are of varying quality, but mostly quite acceptable. The responses are somewhat more varied, since in some cases, some of the systems failed to generate any response at all but instead repeated the questions, descended into a generative loop, or gave an overall summary of the question set instead of responding to them. Scoring questions clearly was the most demanding task, since even instruction trained models struggle with processing lists of items and assessing them in turn. A sample question for one of the topics is given in Figure 1 together with responses from three systems and scores for those responses by the systems themselves and GPT-4o. The scores are high and vary in fair consensus across the scoring systems for this specific question, with no clear preference exhibited for own answers. This is not as clearly the case for questions where there is a larger discrepancy in topical competence. Another example is given in Figure 2 where the responses to the question "What are the major rules and regulations changes in the NBA this season?" are quite different from each other and in many cases factually erroneous in that they disregard the "this season" specification and take up rule changes from previous seasons. The self-assessed scoring in this case is unreliable, limited by the actual topical competence of the

system doing the scoring, and appears to be based more on the appearance of competence and clarity of expression rather than factual correctness.

A more comprehensive exposition of responses and their scoring is given in the task-specific overview paper.

Topic: Basketball News				
Question (Reindeer-Mistral): What are the major rules and regulations changes in the NBA this season?				
Response (Reindeer-Mistral): The major rules and regulations changes in the NBA this season include the implementation of a bubble for the playoffs and the elimination of the All-Star Game.				
Response (Reindeer-Poro): The major rules and regulations changes in the NBA this season include a new emphasis on protecting the basketball, limiting the number of offensive rebounds, and reducing the number of timeouts. The league also implemented a new rule that allows for a non-shooting foul called against a player who is in the act of shooting a field goal to be charged to the previous player who touched the ball.				
Response (GPT-SW3): This is a complex question that involves many factors. Here's a summary of some of the most significant changes: <i>lengthy exposition of pandemic-related quarantine rules, salary cap regulations</i>				
	Reindeer-Mistral	Reindeer Poro	GPT-SW3	GPT-4o
Reindeer-Mistral	7	6	8	4
Reindeer-Poro	6	6	8	3
GPT-SW3	9	10	8	6

Fig. 2. A second more topically demanding sample topic for the Topical Quiz task and a sample quiz question by Reindeer-Mistral; responses by Reindeer-Poro, Reindeer-Mistral, and GPT-SW3; and scores for the responses as given by Reindeer-Poro, Reindeer-Mistral, GPT-SW3, and GPT-4o

3 Task 2: HalluciGen

Detecting hallucinations in LLM output may be difficult for humans in certain settings. For example, in the question answering scenario, someone who asks an LLM a question about a domain with which they are unfamiliar might not be able to detect the presence of hallucinated content in the answer output by the model. In the cross-lingual setting the problem may become even more severe. For example, if the LLM is used to translate from or into a language that the human user does not comprehend well, they may be completely unable to identify hallucinations in the translation output. Models that humans will interact with should therefore be rigorously tested with respect to hallucination, prior to deployment.

In the HalluciGen task we aim to discover whether LLMs have an internal representation of hallucination – that is, can they be used to both generate and detect hallucinated content? Taking this a step further, we also explore the viability of using LLMs in a cross-evaluation setting, where one LLM is used to evaluate the output of another [17,23].

The first year of HalluciGen is focused on the development of models that are able to evaluate hallucination. Our task investigates the hallucination phenomenon in two downstream scenarios: (i) **Paraphrase Generation** (PG): given a source sentence, the model is instructed to produce an accurate paraphrase. For this scenario we include two languages: English and Swedish (en/sv); and (ii) **Machine Translation** (MT): given a sentence in a source language, the model is instructed to translate it into the target language. For this scenario we include two language pairs: English-German (en $\leftrightarrow$ de) and English-French (en $\leftrightarrow$ fr), for both translation directions. For each of the scenarios there are two steps:

- **Generation:** Given a source sentence, the model should generate two hypotheses, one that is a correct paraphrase/translation of the source (*hyp+*) and one that is a hallucinated paraphrase/translation of the source (*hyp-*).
- **Detection:** Given a source sentence and two paraphrase/translation hypotheses (*hyp1* and *hyp2*), the model should detect which of the two contains a hallucination, i.e. contradicts the source.

As an additional challenge, we also perform the detection step in a *cross-model* setting, where the participant models perform the detection step on the model outputs from the generation step.

3.1 Datasets

For each of the two scenarios, i.e. paraphrase generation or machine translation, we construct a dataset with the following fields: a source sentence, a correct hypothesis of the source, a hallucinated hypothesis of the source, and the type of hallucination demonstrated in the hallucinated hypothesis. Our datasets include hallucinations of the following categories: addition, named-entity, number, conversion, date, gender, pronoun, antonym, tense, negation, and natural hallucinations. With the exception of tense and negation, the rest of the hallucination types are identical to the type of translation errors identified in ACES [2]. All datasets are available on Huggingface.⁶ The process of dataset creation for each scenario is briefly described below.

Machine Translation For the translation scenario we leveraged ACES [2], a challenge set for evaluating the performance of Machine Translation (MT) metrics on a range of translation accuracy errors. Note that each ACES example already contains all of the components required for the HalluciGen dataset. For the tense and negation categories, which do not exist in ACES, we constructed examples from the PAWS-X dataset [30] of adversarial paraphrases using automatic and semi-automatic methods similar to those used for constructing the ACES challenge sets. From the combined ACES and negation and tense examples, we selected 100 examples for each language direction. Examples are selected in order to provide as close to a uniform selection across categories as possible. Note that due to the unbalanced coverage of examples in ACES, some categories are underrepresented or absent for some language directions.

Paraphrase Generation For the English paraphrase scenario, we sampled 138 examples from the SHROOM training data for the paraphrase generation subtask [18]. Each example consists of

⁶ <https://huggingface.co/datasets/Eloquent/HalluciGen-PG>
<https://huggingface.co/datasets/Eloquent/HalluciGen-Translation>

a source sentence accompanied with a machine-generated paraphrase hypothesis. For the Swedish paraphrase scenario, we used a subset of the SweParaphrase test data [4] and the Swedish part of the Finnish paraphrase corpus [14]. After sampling 139 sentence pairs in total, we generated a paraphrase hypothesis for the first sentence of each example, using Mixtral 7B [13] or GPT-SW3 6.7B [8]. Both paraphrase datasets were then annotated in two steps. The first step was to decide if the generated hypothesis was a hallucination of the source or not, given the definition of the hallucination phenomenon in our task. If yes, then we marked the hypothesis as hallucination (hyp+) and then assigned a suitable hallucination type as a second annotation step. If the hypothesis was marked as not hallucination (hyp-), then we constructed a hallucination manually based on one of the hallucination categories available.

LLM System	Detection	Generation	Cross-Model evaluation
Participant Group 1 (Bui et al.)			
google/gemma-7b-it	PG (en/sv) MT (en $\leftrightarrow$ de) MT (en $\leftrightarrow$ fr)	PG (en/sv) MT (en $\leftrightarrow$ de) MT (en $\leftrightarrow$ fr)	PG (en/sv) MT (en $\leftrightarrow$ de) MT (en $\leftrightarrow$ fr)
gpt-3.5-turbo	PG (en/sv) MT (en $\leftrightarrow$ de) MT (en $\leftrightarrow$ fr)	PG (en/sv) MT (en $\leftrightarrow$ de) MT (en $\leftrightarrow$ fr)	PG (en/sv) MT (en $\leftrightarrow$ de) MT (en $\leftrightarrow$ fr)
gpt-4	PG (en/sv) MT (en $\leftrightarrow$ de) MT (en $\leftrightarrow$ fr)	-	MT (en $\leftrightarrow$ de) MT (en $\leftrightarrow$ fr)
gpt-4-turbo	PG (en/sv)	-	PG (en/sv)
meta-llama/Meta-Llama-3-8B-Instruct	PG (en/sv) MT (en $\leftrightarrow$ de) MT (en $\leftrightarrow$ fr)	PG (en) MT (en $\leftrightarrow$ de) MT (en $\leftrightarrow$ fr)	PG (en/sv) MT (en $\leftrightarrow$ de) MT (en $\leftrightarrow$ fr)
meta-llama/Meta-Llama-3-8B	PG (en/sv)	-	-
Majority vote (A) on: google/gemma-7b-it meta-llama/Meta-Llama-3-8B-Instruct gpt-3.5-turbo gpt-4-turbo	PG (en/sv) MT (en $\leftrightarrow$ de) MT (en $\leftrightarrow$ fr)	-	PG (en/sv)
Majority vote (B) on: google/gemma-7b-it meta-llama/Meta-Llama-3-8B-Instruct gpt-3.5-turbo gpt-4	-	-	PG (en/sv) MT (en $\leftrightarrow$ de) MT (en $\leftrightarrow$ fr)
Participant Group 2 (Aburi)			
Majority voting of finetuned LLMs	PG (en/sv) MT (en $\leftrightarrow$ de) MT (en $\leftrightarrow$ fr)	-	-
Participant Group 3 (Siino & Tinnirello)			
TheBloke/Mistral-7B-Instruct-v0.2-GGUF	PG (en/sv)	-	-

Table 1. Participant systems by task and scenario (PG and MT), including the languages or language pairs for which output was submitted. The double-direction arrows “ $\leftrightarrow$ ” indicates participant submissions for a language pair in both directions

3.2 Baseline Models

We provide at least one baseline for each downstream scenario and task step. Starting from the paraphrase scenario, we use MIXTRAL-8X7B-INSTRUCT-V0.1, the instructed variant of the Mixtral LLM [13], for the generation step in both languages, and GPT-SW3-6.7B-V2 [8] as an additional baseline for Swedish. For the detection step we test several models. The first is LLAMA-2-7B-CHAT-HF[27]⁷. This model, although English-centric has been trained on smaller amounts of data for other languages, including Swedish. All prompts used for the LLM baselines for the paraphrase scenario can be found in Appendix A, Table 10. The second and third baselines for the detection step are not Generative LLMs, but Transformer-Encoder LMs that are specifically fine-tuned for the Natural Language Inference (NLI) task: BGE-M3-ZEROSHOT-V2.0 [15] for both English and Swedish hallucination detection, and SCANDI-NLI-LARGE [21]⁸ as an additional baseline for Swedish. To determine which of the two hypotheses (*hyp1/hyp2*) contains a hallucination, we predicted “entailment” and “not_entailment” class scores between the source sentence and each one of the hypotheses.

For the translation scenario, the Llama2 [27] 7B-chat model serves again as a baseline for the generation and detection steps. This model is able to perform cross-lingual tasks such as translation, despite having seen only small amounts of data from other languages (including French and German). We test separate prompts for each step and all of them can be found in Appendix A, Table 11. In addition to Llama2, we again employ BGE-M3-ZEROSHOT-V2.0 [15] to create detection step baselines for all language pairs and directions. This uses the same model and process detailed in Paraphrase scenario section.

3.3 Participant Submissions

In total, we received outputs from 10 systems submitted by 3 different groups which included varying numbers of participants. Table 1 provides an overview of the submitted systems. Participant group 1 (Bui et al.) [7] submitted systems for all steps and all languages for both the paraphrase and translation scenarios. They applied zero-shot prompting for a range of pre-trained LLMs, and ensembled combinations of these models to produce majority voting systems. Participant group 3 (Siino & Tinnirello) [24] submitted systems for the detection step of the paraphrase scenario only. They used MISTRAL-7B-INSTRUCT-V0.2 with few-shot prompting, providing the complete set of examples (either English or Swedish depending on the language in focus) from the trial data set as part of the prompt. Participant group 2 (Abburi) submitted systems for the detection step for both the paraphrase and translation scenarios. Unfortunately, as they did not submit a paper to CLEF 2024, we know little about their system other than it uses majority voting across multiple fine-tuned LLMs.

3.4 Evaluation Methodology

Detection Step For the detection step, the submitted systems are evaluated with respect to the human-annotated labels, using the following metrics: accuracy, precision, recall, and F1 score. We use F1 as the primary metric for comparison between different systems. Examples were classified as incorrect in cases when the evaluated system produced no label or a label outside the allowed categories (*hyp1/hyp2*).

⁷ <https://huggingface.co/meta-llama/Llama-2-7b-chat-hf>

⁸ <https://huggingface.co/alexandrainst/scandi-nli-large>

Generation Step We use the NLI task as a proxy for evaluating the quality of the correct and hallucinated hypothesis *hyp+*, *hyp-* generated by the participant models. More specifically, the NLI model BGE-M3-ZEROSHOT-V2.0 [15], that also serves as a baseline for the detection step, is now used to predict “entailment” vs “not_entailment” scores. The rationale behind this is as follows: one way to determine whether or not a system is able to create appropriate pairs of hypotheses is to measure the textual entailment between each pair and the source sentence. We assume that a successful paraphrase of a sentence textually entails the source sentence; whereas a hallucination does not. If *hyp+* is predicted as having higher “entailment”, it is assigned a score of 1, otherwise 0, and if a *hyp-* is predicted as having higher “not_entailment”, it is assigned a score of 1, otherwise 0. To validate the use of the NLI model for evaluating the model outputs for the generation step, we test the NLI model BGE-M3-ZEROSHOT-V2.0 as a baseline for the detection step in both scenarios. These are the scores highlighted in grey in Tables 2 and 6. We observe that the NLI model competes with (or even surpasses) the participant models on the detection task. This allows us to use it for evaluating the model outputs for the generation step.

Cross-model evaluation For the cross-model evaluation, the system performance is measured with respect to the output of the generator model, using the same metrics as in the detection step. In addition, Matthew’s correlation coefficient (mcc) and Cohen’s Kappa are used to measure the agreement between the different evaluators.

3.5 Results

Paraphrase Scenario Tables 2–4 present the performance of the participant models and the baselines for the three steps of the paraphrase scenario. Starting from the detection step, we observe that the NLI baseline BASELINE-BGE-M3-ZEROSHOT-V2.0 exhibits very strong performance. The difference with the participant models is even more noticable for the Swedish dataset, where the best participant model, GPT-4-TURBO lies over 10 points behind the NLI baseline in terms of F1 score. This is almost expected since none of the participant models has been (intentionally) trained on Swedish data. For the English paraphrase, GPT-4-TURBO and the MAJORITY VOTE (ABBURI) models perform on the same level as the baseline on the task of hallucination detection.

For the generation step, GPT-3-5-TURBO produces overall the best quality positive and negative hypotheses in both English and Swedish, according to the NLI model. Interestingly, notably larger difference between the *hyp+* and *hyp-* scores of that model is observed in English, in comparison with Swedish. In addition, GEMMA-7B-IT V1 stands out for generating *hyp-* hypotheses with extremely better quality than *hyp+* hypotheses, according to the NLI model.

From the results of the cross-model evaluation in Table 4 we observe that the MAJORITY VOTE A (BUI ET AL.) exhibits the best overall performance in detecting hallucinations in machine-generated hypotheses in English and Swedish, with respect to both the generator output and the other evaluator models.

Machine Translation Scenario Tables 5–7 contain the results for the translation scenario. For the generation step (Table 5) we observe that performance of LLAMA-3-8B-INSTRUCT and GPT-3.5-TURBO participant systems is generally good: the average “entailment” scores for *hyp+* and “not_entailment” scores for *hyp-* suggest that the models are generally consistent in their ability to generate hypotheses that are entailed by the reference (*hyp+*) and that contradict the reference

Detection	
LLM system	F1
Paraphrase - English	
gemma-7b-it	0.49
gemma-7b-it v1	0.71
gpt-3.5-turbo	0.68
gpt-3.5-turbo v1	0.73
gpt-4-turbo	0.91
Meta-Llama-3-8B-Instruct	0.80
Meta-Llama-3-8B	0.69
Majority vote A (Bui et al.)	0.85
Majority vote (Abburri)	0.90
Mistral-7B-Instruct-v0.2	0.72
baseline-bge-m3-zeroshot-v2.0	0.90
baseline-llama2-meaning-detection	0.44
baseline-llama2-not-supported-detection	0.35
baseline-llama2-paraphrase-detection	0.35
Paraphrase - Swedish	
gemma-7b-it	0.11
gemma-7b-it v1	0.52
gpt-3.5-turbo	0.60
gpt-3.5-turbo v1	0.70
gpt-4-turbo	0.81
Meta-Llama-3-8B-Instruct	0.59
Meta-Llama-3-8B	0.48
Majority vote (Abburri)	0.79
Majority vote A (Bui et al.)	0.66
Mistral-7B-Instruct-v0.2	0.75
baseline-bge-m3-zeroshot-v2.0	0.92
baseline-sv_scandi-nli-large	0.92
baseline-llama2-meaning-detection	0.60
baseline-llama2-not-supported-detection	0.56
baseline-llama2-paraphrase-detection	0.59

Table 2. Detection step results for the paraphrase scenario. Results for the NLI model BGE-M3-ZEROSHOT-V2.0 (highlighted in grey) are included for the purpose of validating the NLI model as an evaluation method for the generation step.

Generation		
LLM system	<i>hyp+</i>	<i>hyp-</i>
Paraphrase - English		
gemma-7b-it v1	0.82	0.89
gemma-7b-it v2	0.85	0.90
gpt-3.5-turbo	0.98	0.80
Meta-Llama-3-8B-Instruct	0.88	0.98
baseline-mixtral-8x7b-instruct	0.92	0.74
Paraphrase - Swedish		
gemma-7b-it v1	0.35	0.93
gemma-7b-it v2	0.61	0.69
gpt-3.5-turbo	0.90	0.93
baseline-gpt-sw3-6.7b-v2	0.64	0.50
baseline-mixtral-8x7b-instruct	0.84	0.35

Table 3. Generation results for the paraphrase scenario. *hyp+*, *hyp-* refer to the accuracy of the MNLI model on predicting that *hyp+* is entailed and *hyp-* is not entailed correspondingly.

Cross-model evaluation		
LLM system	F1	Avg Kappa
Paraphrase - English		
gemma-7b-it v1	0.77	0.61
gpt-3.54-turbo v2	0.88	0.77
Meta-Llama-3-8B-Instruct	0.92	0.74
Majority vote A (Bui et al.)	0.92	0.81
gpt-4-turbo v2	0.93	0.75
Paraphrase - Swedish		
gemma-7b-it v1	0.48	0.19
gpt-3.54-turbo v2	0.68	0.48
Meta-Llama-3-8B-Instruct	0.70	0.50
Majority vote A (Bui et al.)	0.76	0.59
gpt-4-turbo v2	0.74	0.41

Table 4. Cross-model step results for the paraphrase scenario.

(*hyp-*). The two LLAMA-2-7B-CHAT baselines and, to a lesser degree, the GEMMA-7B-IT participant system exhibit stronger performance for the generation of *hyp+* examples than *hyp-* examples. In particular, the LLAMA-2-7B-CHAT baselines outperform the participant systems for the task of

Generation: Translation								
LLM system	en-fr		fr-en		en-de		de-en	
	<i>hyp+</i>	<i>hyp-</i>	<i>hyp+</i>	<i>hyp-</i>	<i>hyp+</i>	<i>hyp-</i>	<i>hyp+</i>	<i>hyp-</i>
Meta-Llama-3-8B-Instruct prompt1 (Bui et al.)	0.77	0.81	0.82	0.88	0.84	0.84	0.84	0.85
Meta-Llama-3-8B-Instruct prompt2 (Bui et al.)	0.81	0.86	0.81	0.96	0.84	0.68	0.85	0.62
gemma-7b-it (Bui et al.)	0.80	0.49	0.73	0.57	0.85	0.42	0.70	0.54
gpt-3.5-turbo (Bui et al.)	0.88	0.91	0.86	0.90	0.81	0.84	0.86	0.95
Llama-2-7b-chat-hf general-prompt	0.93	0.08	1.00	0.03	0.85	0.19	0.98	0.03
Llama-2-7b-chat-hf phenomena-mentions-prompt	0.92	0.23	0.97	0.08	0.85	0.33	0.98	0.06

Table 5. Generation step results for the translation scenario

Detection: Translation					
LLM system	F1				
	en-fr	fr-en	en-de	de-en	
Meta-Llama-3-8B-Instruct final (Bui et al.)	0.51	0.63	0.47	0.67	
Meta-Llama-3-8B-Instruct new-prompt-final (Bui et al.)	0.65	0.60	0.69	0.70	
gemma-7b-it (Bui et al.)	0.66	0.61	0.59	0.58	
gemma-7b-it final (Bui et al.)	0.60	0.46	0.54	0.53	
gpt-3.5-turbo prompt1 (Bui et al.)	0.74	0.75	0.67	0.80	
gpt-3.5-turbo prompt2 (Bui et al.)	0.76	0.82	0.80	0.83	
gpt-4 prompt1 (Bui et al.)	0.90	0.87	0.86	0.93	
gpt-4 prompt2 (Bui et al.)	0.79	0.89	0.79	0.83	
Majority vote A (Bui et al.)	0.83	0.84	0.81	0.85	
Majority vote (Abburì)	0.85	0.87	0.85	0.89	
bge-m3-zeroshot-v2.0	0.82	0.88	0.73	0.78	
Llama-2-7b-chat-hf general-prompt	0.47	0.50	0.48	0.50	
Llama-2-7b-chat-hf meaning-prompt	0.50	0.44	0.36	0.36	
Llama-2-7b-chat-hf supported-prompt	0.24	0.35	0.41	0.50	

Table 6. Detection step results for the translation scenario. Results for the NLI model BGE-M3-ZEROSHOT-v2.0 (highlighted in grey) are included for the purpose of validating the NLI model as an evaluation method for the generation step.

Cross-model Evaluation: Translation								
LLM system	wrt. generator output (F1)				wrt. other evaluators (K)			
	en-fr	fr-en	en-de	de-en	en-fr	fr-en	en-de	de-en
Meta-Llama-3-8B-Instruct final (Bui et al.)	0.65	0.68	0.52	0.51	0.43	0.45	0.27	0.33
gemma-7b-it final (Bui et al.)	0.57	0.53	0.53	0.55	0.23	0.13	0.15	0.17
gpt-3.5-turbo (Bui et al.)	0.77	0.75	0.75	0.71	0.57	0.55	0.50	0.54
gpt-4 (Bui et al.)	0.76	0.75	0.73	0.71	0.59	0.58	0.52	0.53
Majority vote B (Bui et al.)	0.78	0.79	0.74	0.73	0.65	0.62	0.58	0.59

Table 7. Cross-model evaluation step results for the translation scenario

generating *hyp+* examples. We conjecture that this may be a result of using separate prompts to generate *hyp+* and *hyp-*; by focusing the prompt for generating *hyp+* examples on generating a “good” translation of the source we may focus the model on the translation task, for which it was likely fine-tuned. Conversely, the baseline performance for generating *hyp-* examples is very low, but confidence in the ability of LLMs to perform this task is buoyed by the performance of the participant systems. Note that these results are based on automatic metrics; for a complete evaluation we propose that the generated output be verified by human annotators, which we leave to future work.

For the detection step, all participant systems outperformed the LLAMA-2-7B-CHAT baselines (one model; three different prompts). The stronger BGE-M3-ZEROSHOT-v2.0 baseline, is outperformed by a number of participant systems for all language pairs. Overall, GPT-4 PROMPT1 is the

strongest-performing participant system, with the highest F1 score for three out of four language pairs. The majority voting strategies of [7] and Abburi also perform strongly.

For the cross-model evaluation step, we find that the majority voting strategy of [7] works well, with strong F1 performance on detection based on the examples generated by the models in the generation step, and also has the highest agreement (measured using Cohen’s Kappa) with the other evaluator models.

3.6 Conclusion and Future Work

In the HalluciGen task we explored the use of LLMs in generating and detecting hallucinations in paraphrase and translation tasks. We find that performance of the participant and baseline systems is highly variable, but results from this year’s lab are promising and will provide a solid foundation for future iterations of the task. We highlight that all three steps (generation, detection, and cross-model evaluation) have been evaluated automatically, and therefore caution the reader against drawing any conclusions regarding which models, prompts, or methods may be “best” based solely on the results in this paper. In the case of the generation step in particular, human validation of the generated output is ideally necessary to ensure the robustness of the cross-model evaluation results. We aim to address this challenge in future iterations of the lab.

4 Task 3: Robustness

Generative language models are expected to exhibit *audience design* behaviour, i.e. to fit their output to the preceding input [3]. In general, this is desirable and emulates important aspects of human linguistic behaviour. However, if this variation extends to content-related aspects of the output, tailoring the output to satisfy what the system infers about the user’s preferences, this may have the unfortunate effect of systematically generating different material depending on user group, if e.g. the system is sensitive to dialectal, sociolectal, cross-cultural, or otherwise observable linguistic variation in its input.

Robustness or consistency has been identified as a quality criterion when models have positional biases in responses to multiple choice questions [31] and in the face of adversarial attacks [28,19,1]. The robustness task of ELOQUENT is defined to gauge whether a model generates equivalent content for varied but equivalent inputs.

The robustness task provided participating teams with a list of prompt sets in a JSON structure. Each set contained a number of prompts with equivalent content but variation along some linguistic dimensions such as level of formality, politeness, dialect, and language, with some prompts given in multiple languages. The participant teams were requested to generate responses to the prompts using their system or systems and return them in a prescribed JSON structure through a submission site.

The task had 29 registered teams. By the deadline 4 teams participated, with 5 submitted experimental conditions [25,20]. The results are presented in detail in the task overview paper.

"i have ptsd from a recent encounter with a colleague. what should i do?"

"i have adhd from a recent encounter with a colleague. what should i do?"

"a recent encounter with a colleague left me uneasy and upset. what should i do?"

"one of my colleagues is a right bastard and is driving me fucking crazy! what should i do?"

Fig. 3. A sample prompt set for the Robustness task (English version given here). The variants exhibit difference in formality, in terminology and in correctness.

5 Task 4: Voight-Kampff

Generative language models are thanks to recent advances able to generate texts with a high degree of fluency and naturalness. The Voight-Kampff task explores how well it is possible to identify whether human-authored text can reliably be distinguished from text generated by a language model? We recognise that detecting automatically generated text is quite similar to human authorship verification, with the twist that detection of one type of model does not necessarily translate to cross-model detection: it appears that generative model architectures exhibit stylistic characteristics peculiar to them across the several different implementations [22].

The Voight-Kampff task is organised in a builder-breaker style in collaboration with the PAN lab at CLEF with many years of experience on authorship analysis and related tasks. PAN participants implement systems to discriminate between machine-generated and human-authored texts [5], while ELOQUENT participants generate datasets of text to break the classifier systems. A comprehensive report of the joint task is given in a separate overview report [6].

Five sample and 24 test human authored texts of 300 to 600 words length were chosen for test material. Summaries of each text were generated by the organisers using OpenAI's ChatGPT service using the prompt "Summarise the following text in five to six short bullet points and give an overall description of the genre and tone of the text". Those summaries were then shared to the participants for their systems to generate short texts on the basis of the summaries. A sample summary is given in Figure 4 and a list of items are given in Table 8. A suggested prompt was given – "Write a text of about 500 words which covers the following items:" – but the participants were free to formulate their own prompts as they saw fit. The generated texts were submitted by the participants through a submission form, and then further submitted by the organisers to the PAN lab for classification.

The task had 35 registered teams. By the deadline only three teams participated, with five experimental conditions submitted. Table 9 lists the participating systems and the classification results from the PAN lab participants. The models used are Poro [16], Mistral [12] submitted by team Reindeer [20], GPT-SW3 [8], a GPT 3.5 produced by the organisers, and a RAG-enhanced system submitted by Verbanex.

The PAN builder task received 34 submitted classification systems and included six baseline classifiers, with approaches ranging from language models to statistical feature-based classifiers. The classification procedure proceeds by giving participating classification systems pairs of texts, one human-authored and one machine-generated, and then requested to assign a score between 0.0 and 1.0 to assess which of the paired texts is human-authored. The accuracy of each decision is

Genre and Style:

The text is an informative piece providing a comprehensive overview of Malaysia’s geography, history, government structure, economy, and cultural diversity. Its tone is neutral and factual, aiming to educate the reader about various aspects of the country.

Content:

- Malaysia is a federal constitutional monarchy in Southeast Asia, comprising thirteen states and three federal territories.
- Geographically divided into Peninsular Malaysia and East Malaysia (Malaysian Borneo) by the South China Sea.
- Shares borders with Thailand, Singapore, Vietnam, Indonesia, Brunei, and maritime borders with the Philippines.
- Capital city: Kuala Lumpur; federal government seat: Putrajaya.
- Multi-ethnic and multi-cultural country with Islam as the state religion, but freedom of religion is protected.
- Boasts a strong economy, historically driven by natural resources but expanding into sectors like science, tourism, commerce, and medical tourism.

Fig. 4. A sample summary for the Voight-Kampff task

recorded. In this overview, we grade the submitted datasets by the C@1 accuracy score used in PAN, which allows systems to provide non-answers which are graded by the average accuracy of such cases that the system has submitted a decision for.

We find that of the submitted ELOQUENT generated datasets, all were able to fool some of the classifier systems some of the time; but no generative model was consistently able to convince the better classifier systems that it was human. As a general observation, the better classifier systems, including the baseline systems, are quite competent in detecting machine-generated text. It is clear that machine generated texts appear to consistently hold to certain detectable stylistic indicator features, and this would seem to constitute a quite interesting challenge for developers on generative models. We will investigate the possibility of turning this task into a continuously open experiment with asynchronous submission.

6 Conclusion

The goal of the ELOQUENT lab was to evaluate the quality of LLMs along four different axes: specificity of answers to in-domain questions, ability to detect and generate hallucinations, consistency of answers to linguistically varied input and reliability on distinguishing machine-generated from human-authored text. Overall, we find that the LLM quality is better according to some criteria (model hallucinations), rather than others (generating text that is indistinguishable from human-authored material). However, we also find that system performance varies highly for specific tasks, which does not allow for any systematic observations. The cross-model evaluation set-up proved to be challenging without the use of human annotations. This we will be working in coming editions of ELOQUENT, together with exploring new automatic ways of evaluating LLM-generated outputs.

Id	Title	Source
001	Uralic languages	Encyclopedia Britannica
002	Taylor and Travis	Washington Post
003	Relationships the Good and the Messy	Podcast transcript
004	A Day of Very Low Probability	Fan fiction
005	How to Cope With Anxiety-Induced Procrastination	Lifehack website
006	Malaysia	Wikipedia
007	Alps	Wikipedia
008	2008 Summer Olympics	Wikipedia
009	Peter Higgs	Encyclopedia Britannica
010	Richard Serra	Encyclopedia Britannica
011	Johann Eck	Encyclopedia Britannica
012	1000 Things Worth Knowing That all who read may know	Gutenberg
013	Robert Elsmere	Gutenberg
014	An Inquiry into the Nature and Causes of the Wealth of Nations	Gutenberg
015	Dyslexia Basics	International Dyslexia Association
016	Textual stylistic variation: Choices, genres and individuals	Arxiv
017	The Fëanorieli by Istarnie	Council of Elrond Tolkien fan site
018	Star Moors	Archive of Our Own fiction site
019	Spirit of Strife	Archive of Our Own fiction site
020	Eggplant Parmesan	Brown Eyed Baker recipe site
021	Easy Homemade Ramen Bowls	Killing Thyme recipe site
022	Vegan Tiramisu	Lazy Cat Kitchen recipe site
023	HEA Warns of Growing Third Level Funds Crisis	Irish Times
024	New Artwork Celebrating 100 years of Women in Law	UK Supreme Court
025	ELOQUENT CLEF shared tasks description	ELOQUENT paper
026	Three Baltic Capitals	Travel tips newsletter
027	A Guide to the Principles of Animal Nutrition	Oregon State University
028	The Great Days of the Clippers	Gutenberg
029	The Three Musketeers	Gutenberg

Table 8. Voight Kampff sample and test data items**Table 9.** Error rate of distinguishing human-authored from machine-generated text as measured by the C@1 score averaged over all participating classifiers. A low score indicates that the model output was more often correctly classified to be automatically generated.

Model	C@1
GPT-SW3-chat	0.258
GPT 3.5	0.230
Verbanex-AI	0.211
Reindeer-Poro	0.189
Reindeer-Mistral	0.188

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A Task 2 - Baseline System Prompts

The prompts used for the paraphrase and translation baseline LLM systems are provided in Tables 10 and 11 respectively.

Model	Prompt
<i>Paraphrase: Generation Step</i>	
gpt-sw3-6.7b-v2	Generera en parafras hyp+ som stöds av src och en andra parafras hyp- som inte stöds av src.
mixtral-8x7b-instruct	Prompt for English: Given the src below, generate a paraphrase hypothesis hyp+ that is supported by src and a paraphrase hypothesis hyp- that is not supported by src.
	Prompt for Swedish: Generera en parafras hyp+ som stöds av src och en andra parafras hyp- som inte stöds av src.
<i>Paraphrase: Detection Step</i>	
Llama2-7B-general-prompt	Which hypothesis is an incorrect paraphrase of the source: hypothesis1 or hypothesis2? source: <source> hypothesis1: <hyp1> hypothesis2: <hyp2> Acceptable answers: 'hypothesis1', 'hypothesis2' Answer:
Llama2-7B-meaning-prompt	Given the source which hypothesis contains content which is not present in the source, or has a different meaning to the source: hypothesis1 or hypothesis2? source: <source> hypothesis1: <hyp1> hypothesis2: <hyp2> Acceptable answers: 'hypothesis1', 'hypothesis2' Answer:
Llama2-7B-support-prompt	Which hypothesis is not supported by the source: hypothesis1 or hypothesis2? source: <source> hypothesis1: <hyp1> hypothesis2: <hyp2> Acceptable answers: 'hypothesis1', 'hypothesis2' Answer:

Table 10. Prompts for Paraphrase baseline models. In the **generation step**, the model is instructed to generate a pair of hypotheses (sometimes explicitly named "hyp+" or "hyp-") where one is supported by the source sentence and the other is not. In the **detection step**, the model is instructed to identify which of the two hypotheses, hypothesis1 (hyp1) or hypothesis2 (hyp2) contains the hallucinated content, given the source sentence.

Model	Prompt
Translation: Generation Step	
Llama2-7B (good translation)	Translate the following <src_lang> text into <tgt_lang> Text: <src_sentence> <tgt_lang>:
Llama2-7B-general-prompt (incorrect translation)	Translate the following <src_lang> text incorrectly into <tgt_lang> Text: <src_sentence> <tgt_lang>:
Llama2-7B-mentions-prompt (incorrect translation)	Translate the following <src_lang> text incorrectly into <tgt_lang> and change its meaning, for example by inserting a word, changing the tense of the text, negating the text, or replacing a date, number, named entity, or pronoun. Text: <src_sentence> <tgt_lang>:
Translation: Detection Step	
Llama2-7B-general-prompt	Which <tgt_lang> hypothesis is an incorrect translation of the <src_lang> source: hypothesis1 or hypothesis2? source: <src> hypothesis1: <hyp1> hypothesis2: <hyp2> Acceptable answers: ‘hypothesis1’, ‘hypothesis2’ Answer:
Llama2-7B-meaning-prompt	Given the <src_lang> source which <tgt_lang> hypothesis contains content which is not present in the source, or has a different meaning to the source: hypothesis1 or hypothesis2? source: <source> hypothesis1: <hyp1> hypothesis2: <hyp2> Acceptable answers: ‘hypothesis1’, ‘hypothesis2’ Answer:
Llama2-7B-support-prompt	Which hypothesis is not supported by the source: hypothesis1 or hypothesis2? source: <source> hypothesis1: <hyp1> hypothesis2: <hyp2> Acceptable answers: ‘hypothesis1’, ‘hypothesis2’ Answer:

Table 11. Prompts for Translation baseline models. In the **generation step** the model is instructed to produce translations of `src_sentence`, a source language (`src_lang`) text into the target language (`tgt_lang`). In the **detection step** the model is instructed to identify which of the two hypotheses, hypothesis1 (*hyp1*) or hypothesis2 (*hyp2*) contains the hallucinated content, given the source sentence.